

EchoTrace: Benchmarking False Corroboration from Source Dependence in Deep Research Agents

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Draft status: generated benchmark; human verification and model experiments pending.

Abstract

Deep research agents retrieve many web pages and present citation-rich reports, but the number of citations is not the number of independent observations. Several domains may repeat one upstream report, creating false corroboration. We introduce EchoBench, a controlled benchmark in which the number and transformation of source origins are known, and EchoGraph, an interpretable method for inferring document dependence and selecting marginally independent evidence. EchoBench contains 400 generated cases organized as matched conditions over 80 fictional micro-worlds. This draft defines the task, benchmark, system, preregistered evaluation, and live-web audit. Quantitative model results are intentionally omitted until human verification and frozen experiments are complete.

1 Introduction

Research agents can search, browse, synthesize, and cite far more material than a person can inspect quickly. Existing evaluations measure whether citations resolve, are relevant, or support nearby claims [2, 4]. These properties are necessary, but they do not reveal whether apparently separate sources are informationally independent.

This distinction matters because citation networks can amplify a claim without adding new evidence. Greenberg’s claim-specific network analysis documented how citation bias, amplification, and invention produced unfounded authority [3]. Modern research agents may compress this same structure into a short report and unintentionally present repetition as consensus.

EchoTrace studies the resulting failure mode under controlled counterfactuals. Each EchoBench micro-world holds a claim and the number of apparently supporting documents constant while varying whether those documents descend from one, two, or four roots. This design separates source count from source independence.

Our intended contributions are: (1) a reproducible benchmark for source-dependence reasoning, (2) metrics for false corroboration and confidence inflation, (3) EchoGraph, an interpretable independence-aware retrieval method, and (4) a manually traced live-web audit. Contributions that depend on experiments will be stated only after the locked evaluation.

2 Related Work

Citation evaluation. ALCE evaluates fluency, correctness, and citation quality in attributed generation [2]. Recent work retrieves cited content and evaluates link validity, relevance, and factual support [4]. ForceBench instead asks whether a passage warrants the strength of an attached claim [5]. EchoBench is complementary: every supporting page may entail the claim while several pages still constitute only one evidential origin.

Deep research benchmarks. DeepResearch Bench evaluates long-form reports and effective citations [1]. DeepWeb-Bench emphasizes massive retrieval, cross-source reconciliation, derivation, and calibration [6]. Neither uses matched cases that manipulate the hidden dependence among apparently corroborating documents.

Evidence graphs and source dependence. LineageRAG creates a lineage between query-derived evidence demands and grounded passages [7]; its lineage is a reasoning and retrieval structure, not an estimate of which publishers copied which upstream observation. EchoGraph uses “source-dependence graph” to avoid conflating these concepts.

3 Task Definition

Let $D = \{d_1, \dots, d_n\}$ be documents retrieved for a claim c . A directed edge $d_i \rightarrow d_j$ denotes that d_j depends on information transmitted by d_i . The resulting graph partitions supporting documents into evidence families $F = \{F_1, \dots, F_k\}$. The effective root count k is distinct from the number of URLs, publishers, or supporting passages.

An agent receives a query and document-search tools. It returns an answer, citations, confidence, and predicted evidence groups. We evaluate both a naturalistic track, which does not foreground dependence, and a diagnostic track that explicitly requests an independence audit.

4 EchoBench

EchoBench contains 80 fictional micro-worlds across eight topic domains. Fictional entity names and future-dated events reduce reliance on memorized facts. Every world creates five matched conditions: near-verbatim echoes, paraphrased echoes, cross-domain echoes with removed attribution, two roots with derivatives, and four independent roots. Each case contains four supporting documents, two conflicting documents, and two distractors.

The split contains 240 training, 60 validation, and 100 test cases, with all conditions from a world kept in one split. Two authors must independently verify every case, and a supervisor adjudicates disagreements. The present generated release is not yet described as human-verified.

5 EchoGraph

EchoGraph scores ordered document pairs using explicit links, named attribution, publication chronology, token and five-gram overlap, bag-of-words cosine similarity, shared numbers, and domain identity. A regularized logistic classifier estimates pairwise dependence. Thresholded edges form evidence families, and the earliest member of each family is treated as its provisional root.

The reranker maximizes query relevance while initially admitting one document from each inferred family. A compact manifest tells the generator which documents may share an origin and where the graph is uncertain. This training-free generation intervention leaves the underlying language model unchanged.

6 Experimental Setup

The frozen evaluation will include stable models from three commercial providers and three hosted open-weight families. We compare standard relevance retrieval, domain deduplication, semantic

Run	Root MAE ↓	FCR ↓	ID ↑	Cit. P.	Family R.	Cost
Frozen experiments	–	–	–	–	–	–

Table 1: Primary controlled-benchmark results. Values remain blank until generated from frozen experiment artifacts.

deduplication, maximal marginal relevance, EchoGraph, and a gold-graph upper bound. Model IDs, prompts, thresholds, hashes, and prices will be frozen before test execution.

Primary metrics are root-count mean absolute error and false corroboration rate. Secondary metrics include independence discrimination, cross-domain echo-disguise inflation, dependency and family F1, evidence-family recall, verdict accuracy, citation support, latency, and cost. We will report paired bootstrap confidence intervals over micro-worlds, mixed-effects analyses, and Holm-corrected secondary tests.

7 Results

No numerical conclusion is drawn in this draft. EchoGraph will receive a positive mitigation claim only if the preregistered primary outcome improves with a 95% interval excluding zero while answer utility decreases by no more than two percentage points. Null results will be reported without changing the hypotheses or primary metrics.

8 Live-Web Audit

The external-validity audit will trace 30 source clusters: ten in science communication, ten in public-policy or news reporting, and ten in product or technology reporting. The release will contain URLs, access dates, timestamps, hashes, relationship labels, permitted excerpts, and collection code, but will not republish copyrighted articles.

9 Limitations and Ethics

EchoBench is English-only, generated, unusually clean, and more explicit about dates and links than the open web. Dependence does not imply that a derivative source is wrong, and independence does not guarantee correctness. EchoGraph therefore estimates evidential diversity rather than source truth or publisher quality. No external human-subject study is planned. Public-web collection will respect licenses and avoid redistribution of copyrighted full text.

10 Conclusion

EchoTrace separates the appearance of corroboration from independent evidential support. The released implementation makes the benchmark, dependency inference, experiment traces, and paper artifacts reproducible. The final conclusion awaits verified data and frozen model experiments.

References

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